

Student: Aryan Sharma Class: 10-B Roll No: 14

Subject: Science - Light & Optics

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Ans 1:

The speed of light in a vacuum is 3×10^8 metres per second or approximately 299,792,458 m/s.

Ans 2:

Refractive index (n) is the ratio of the speed of light in vacuum to the speed of light in the medium.

Formula: $n = c / v$ where c is speed in vacuum and v in medium.

Ans 3 (a):

Using Snells Law: $n_1 \cdot \sin(i) = n_2 \cdot \sin(r)$

$$1.33 \cdot \sin(30) = 1.5 \cdot \sin(r)$$

$$1.33 \cdot 0.5 = 1.5 \cdot \sin(r)$$

$$0.665 = 1.5 \cdot \sin(r)$$

$$\sin(r) = 0.665 / 1.5 = 0.4433$$

$$r = \arcsin(0.4433) = \text{approximately } 26.3 \text{ degrees}$$

Ans 3 (b):

Optical fibres use total internal reflection to transmit data as light pulses over long distances with minimal loss.

Ans 4:

[Ray diagram drawn - concave mirror with object beyond C,

image formed between F and C, real and inverted]